Uncertainty intervals

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Ask which target, assumptions and confidence method make the interval meaningful.

Teach arithmetic, repeated-sampling interpretation and supplied-bound limits with an authoritative method reference.

Keep one habit: Model main units/SE → contrast raw spread and mean uncertainty → locate zero in supplied intervals → assess independent calculation → retain bias/model/practical-value limits.

MAIN LESSON · SECTION 1

Prepare the connected research investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of uncertainty intervals

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Sampling variability. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of sampling variability

Reference example, not the only acceptable wording: Teacher explanation: Sampling variability means different eligible samples can give different estimates even under the same population. Separate this from procedure changes.

Consider finite population values1, 1, 3, 3 with distinct IDs. Enumerate the six unordered samples of size2 without replacement. Their means are1, 2, 2, 2, 2, 3.

Without opening the old lesson, explain one key idea from Experimental units and independence. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of experimental units and independence

Reference example, not the only acceptable wording: Teacher explanation: The experimental unit is the independently assigned or sampled entity. Multiple measurements from one unit are often related.

If methods are compared within each query, queries are paired units. Repeated runs within a query estimate run variation but do not increase the number of independent query contexts.

Without opening the old lesson, explain one key idea from A literature comparison matrix. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of a literature comparison matrix

Reference example, not the only acceptable wording: Teacher explanation: A literature matrix compares studies using aligned columns rather than sorting headline numbers with different meanings.

Record question, data, unit, baseline, metric, split, resources and limitations. Mark incomparable fields rather than forcing a ranking.

Pictured end goal

Explain uncertainty intervals using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Sampling variability

Experimental units and independence

A literature comparison matrix

Compact alternative: two 45-minute sessions for the core explanation and W1–W5 only. The main plan uses three sessions; schedule further time for the complete investigation, project, visual journey and any code examples. An uncertainty interval needs a sampling model and assumptions. A confidence level refers to the repeated-sampling method, not a probability attached to a fixed unknown parameter after observing data.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of uncertainty intervals

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach model-dependent calculation and honest supplied-interval interpretation

> Separate mean uncertainty, individual variation, confidence coverage and practical importance.

Definitions and complete known-sigma contract

An uncertainty interval needs a sampling model and assumptions. A confidence level refers to the repeated-sampling method, not a probability attached to a fixed unknown parameter after observing data.

For this constructed exercise assume independent normal measurements with known population standard deviation 2. A 95% mean interval uses supplied normal critical value 1.96: mean plus/minus 1.96*sigma/sqrt(n). Un**known sigma** requires a different justified method.

Interval item	Complete constructed-exercise contract
Target	Population mean, not the range of individual observations
Sampling assumptions	Independent measurements from a normal population; known positive population standard deviation
Main supplied summaries	Observed mean 10 units; sigma 2 units; n=16

Interval item	Complete constructed-exercise contract
Standard error of sample mean	SE = sigma / sqrt(n)
Supplied critical value	1.96, rounded normal critical value for approximately 95% coverage
Margin and endpoints	Margin =1.96×SE; interval = mean ± margin
Confidence interpretation	Long-run interval-procedure coverage under assumptions; fixed-parameter probability not supplied
Known sigma versus estimated spread	Do not substitute sample SD into this exact known-sigma formula without a justified method
Model contrast	This independent-normal model differs from the previous finite-population without-replacement exercise
Limits	Numeric guards cannot verify independence, normality, known sigma or unbiased sampling; interval does not repair bias or establish practical value/causality

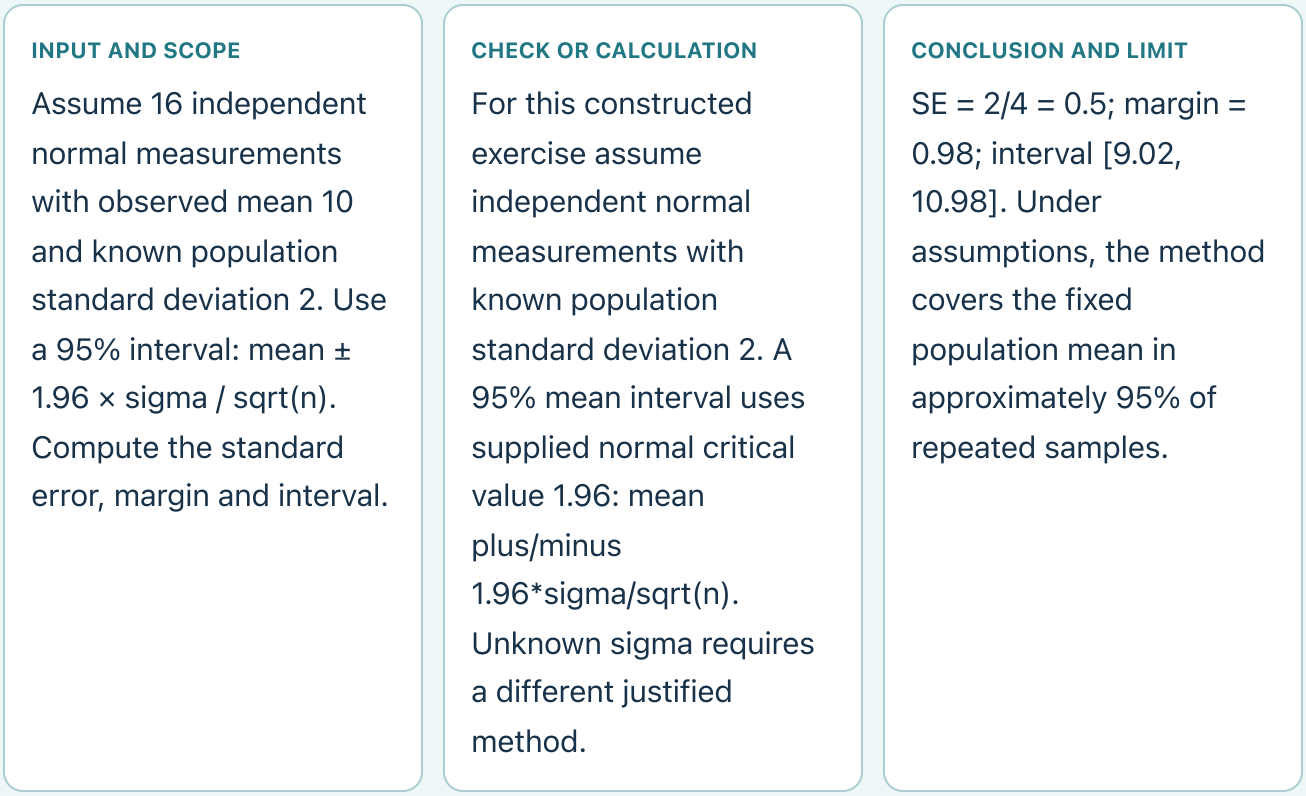
Main arithmetic ledger and checked reasoning

Main calculation step	Substitution	Result and unit
Root sample size	sqrt(16)	4, dimensionless
Standard error	sigma / sqrt(n) =2/4	0.5 measurement units
Margin	1.96 ×0.5	0.98 measurement units
Lower endpoint	10 −0.98	9.02 measurement units
Upper endpoint	10 +0.98	10.98 measurement units
Reported interval	[9.02, 10.98]	For the population mean under stated assumptions
What is not calculated	Raw individual-value coverage or unknown-sigma interval	No actual sampling or coverage experiment

Main: mean 10, known sigma 2 and n=16 give SE 0.5, margin 0.98 and interval [9.02, 10.98] units. Independent normal sampling and known positive population sigma are assumed, not verified. Rounded 1.96 gives about 95% normal-model coverage. The target is the population mean, not 95% of raw observations or a posterior probability for a fixed mean.

Picture: supplied B–A estimate −0.8 seconds with bounds [−1.4, −0.2] lies below zero; changed bounds [−1.4, 0.3] include zero without proving equality. No SE or symmetric fitting method can be inferred from missing raw data. Selected mean 20, sigma 5 and n=25 give SE 1, margin 1.96 and [18.04, 21.96]. Bias, causality and practical utility remain outside the interval guarantee.

Uncertainty intervals — evidence trace



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: SE = 2/4 = 0.5; margin = 0.98; interval [9.02, 10.98]. Under assumptions, the method covers the fixed population mean in approximately 95% of repeated samples. Qualification: A narrow interval can be precisely biased; sampling independence and representativeness must

be justified. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Assume 16 independent normal measurements with observed mean 10 and known population standard deviation 2. Use a 95% interval: mean ± 1.96 × sigma / sqrt(n). Compute the **standard error**, margin and interval.

SE = 2/4 = 0.5; margin = 0.98; interval [9.02, 10.98]. Under assumptions, the method covers the fixed population mean in approximately 95% of repeated samples.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Target, model and sign-interpretation probes

Interval probe	Expected interpretation
Known sigma 2 and n16	SE 0.5; does not mean raw observations have SD 0.5
A 95% confidence method	Approximate long-run coverage under assumptions, not 95% of individual observations
Fixed observed interval	No 95% posterior parameter probability supplied by this frequentist calculation
Unknown sigma	Choose an appropriate justified method; do not silently pretend known
Biased sampling or correlated records	Numeric interval may not have intended coverage
Zero inside a difference interval	No definite sign conclusion from these bounds; not proof of equality
Negative B–A mean interval	B lower mean time under stated convention, not every paired request faster
Supplied asymmetric interval	Keep supplied endpoints; no assumed symmetric-SE reconstruction

Interval probe	Expected interpretation
Finite positive inputs pass code guards	Arithmetic admissible; real model/design assumptions still unverified

Separate supplied time intervals and native zero comparison

Read a supplied uncertainty interval. A fictional study supplies an interval for a mean time difference under its stated sampling model.

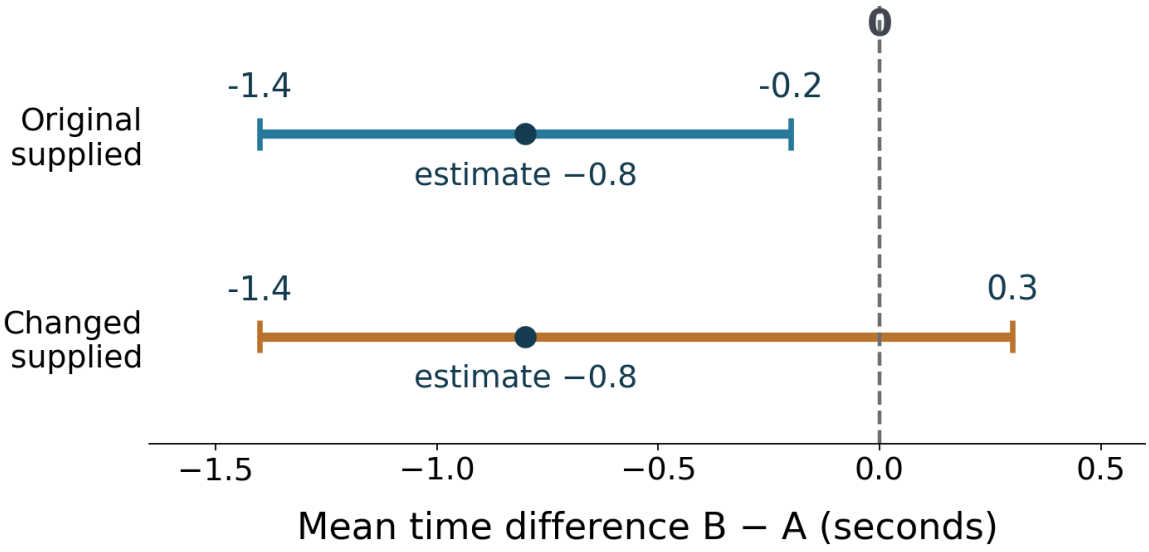
Estimated difference B–A is -0.8 seconds; supplied 95% **confidence interval** is [-1.4,-0.2] seconds. Assume the study's interval method is appropriate.

The picture is a separate time-difference report. Its estimate B–A is –0.8 seconds and its supplied 95% interval is [–1.4, –0.2], assuming the study's method is appropriate. All endpoints are negative, so the interval supports a negative mean-time difference under that method, not a claim that every request is faster.

The changed supplied interval [–1.4, 0.3] includes zero and both negative/positive differences; it does not establish a definite direction or prove equality. Raw data, sample size and fitting method are absent. Do not derive its standard error, force symmetric bounds or apply the main known-sigma formula to it. Practical value and causality need additional evidence.

Does the supplied interval include zero?

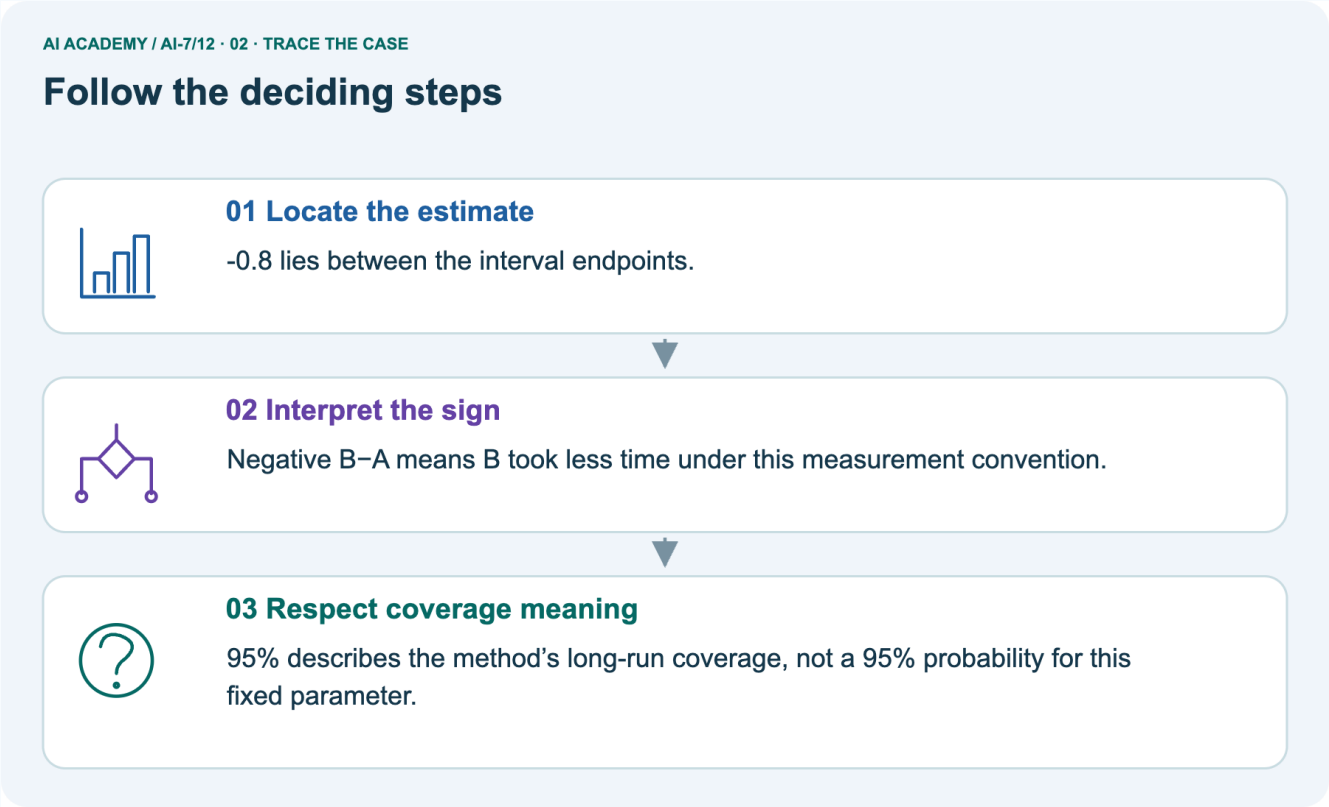
Assume the study’s 95% interval method is appropriate.



Intervals are supplied, not fitted here. Zero inside does not prove equality.

Separate supplied report	Original interval	Changed interval
Point estimate B–A	−0.8 seconds	−0.8 seconds, retained in the scenario
Bounds	[−1.4, −0.2] seconds	[−1.4, 0.3] seconds
Zero relative to bounds	Outside, to the right	Inside
Mean-difference direction	Entire supplied range negative	Both signs and zero included
Valid caution	Not every request’s time or a causal effect	Zero inclusion is not proof of equality
Unavailable ingredients	Raw data and interval fitting details	Still unavailable; no recalculation claimed

Picture trace and zero-including-bound key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Locate the estimate	-0.8 lies between the interval endpoints.
2. Interpret the sign	Negative B–A means B took less time under this measurement convention.
3. Respect coverage meaning	95% describes the method's long-run coverage, not a 95% probability for this fixed parameter.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

An uncertainty interval conveys precision under assumptions. Its endpoints, units, comparison direction and method all matter; the interval is supplied here, not calculated from unavailable raw data.

An interval does not establish practical value or causality, and invalid sampling assumptions can invalidate its interpretation.

Change: Use a supplied interval [-1.4,0.3] instead.

Prediction: It includes zero and positive differences, so both no difference and slower B remain compatible with that interval.

Reason: The new upper endpoint crosses zero.

Primary method reference

Primary method reference: NIST/SEMATECH e-Handbook, "What are confidence intervals?", https://itl.nist.gov/div 898/handbook/prc/section 1/prc 14.htm. It states the normal-mean known-sigma formula and repeated-sampling coverage interpretation; the constructed lesson still requires its declared sampling assumptions.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Standard error	The model-based variation of an estimate
Confidence interval	An interval procedure with declared repeated-sampling coverage
Known sigma	A supplied population standard deviation assumption

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary **input**, record the outcome and state the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
from math import sqrt

def known_sigma_interval(mean, sigma, n, critical=1.96):
    """Assume independent normal sampling and a known population sigma."""
    if type(n) is not int or n < 1 or sigma < 0 or (critical <= 0):
        raise ValueError('interval settings')
```

```
    se = sigma / sqrt(n)
    margin = critical * se
    return (se, mean - margin, mean + margin)

print(known_sigma_interval(10, 2, 16))
```

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Uncertainty intervals

The facts we will use

AI ACADEMY / AI-7/12

Supplied interval for B minus A time

	Lower endpoint (s)	Estimate (s)	Upper endpoint (s)
Original supplied interval	-1.4	-0.8	-0.2
Changed supplied interval	-1.4	not specified	0.3

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/12

Uncertainty intervals

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Locate the estimate

-0.8 lies between the interval endpoints.



2. Interpret the sign

Negative B-A means B took less time under this measurement convention.



3. Respect coverage meaning

95% describes the method's long-run coverage, not a 95% probability for this fixed parameter.

CHANGE ONE THING

Use a supplied interval [-1.4,0.3] instead.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Assume 16 independent normal measurements with observed mean 10 and known population standard deviation 2. Use a 95% interval: mean ± 1.96 × sigma / sqrt(n). Compute the **standard error**, margin and interval. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: SE = 2/4 = 0.5; margin = 0.98; interval [9.02, 10.98]. Under assumptions, the method covers the fixed population mean in approximately 95% of repeated samples.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Uncertainty intervals. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A narrow interval can be precisely biased; sampling independence and representativeness must be justified.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Locate -0.8 and zero relative to the supplied interval [-1.4,-0.2]. Interpret the sign and confidence-method meaning, then describe what changes with interval [-1.4,0.3].

-0.8 lies between the interval endpoints. Negative B–A means B took less time under this measurement convention. 95% describes the method’s long-run coverage, not a 95% probability for this fixed parameter.

Changed case: It includes zero and positive differences, so both no difference and slower B remain compatible with that interval. The new upper endpoint crosses zero.

Changed-case reasoning: It includes zero and positive differences, so both no difference and slower B remain compatible with that interval. The new upper endpoint crosses zero.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main known-sigma calculation yields SE 0.5, margin 0.98 and interval [9.02, 10.98]. Picture difference bounds are supplied and interpreted against zero, not fitted from raw data; zero inclusion does not prove equality. Selected mean 20, known sigma 5 and n=25 give SE 1, margin 1.96 and [18.04, 21.96].

Require independent normal sampling and distinguish long-run method coverage from fixed-parameter probability, bias correction and practical value.

Diagnose the current misconceptions

The worked result establishes every future case: A narrow interval can be precisely biased; sampling independence and representativeness must be justified.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows	Next teaching action
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new research scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Observed mean is 20 from 25 independent normal measurements; population standard deviation is known to be 5. Use the supplied 95% rule mean ± 1.96 × sigma / sqrt(n). Compute the interval and name one assumption.

Teacher explanation: SE 1, margin 1.96, interval [18.04, 21.96]. Require independent normal sampling and **known sigma**; bias is not repaired by the interval.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: An uncertainty interval needs a sampling model and assumptions. A confidence level refers to the repeated-sampling method, not a probability attached to a fixed unknown parameter after observing data. For this constructed exercise assume independent normal measurements with known population standard deviation 2. A 95% mean interval uses supplied normal critical value 1.96: mean plus/minus 1.96*sigma/sqrt(n). Un**known sigma** requires a different justified method.

W2 Trace the worked case

Assume 16 independent normal measurements with observed mean 10 and known population standard deviation 2. Use a 95% interval: mean ± 1.96 × sigma / sqrt(n). Compute the **standard error**, margin and interval. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: SE = 2/4 = 0.5; margin = 0.98; interval [9.02, 10.98]. Under assumptions, the method covers the fixed population mean in approximately 95% of repeated samples.

W3 Audit the limit

A report omits this qualification: A narrow interval can be precisely biased; sampling independence and representativeness must be justified.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A narrow interval can be precisely biased; sampling independence and representativeness must be justified.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A narrow interval can be precisely biased; sampling independence and representativeness must be justified.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: SE = 2/4 = 0.5; margin = 0.98; interval [9.02, 10.98]. Under assumptions, the method covers the fixed population mean in approximately 95% of repeated samples. Qualification: A narrow interval can be precisely biased; sampling independence and representativeness must be justified. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Observed mean is 20 from 25 independent normal measurements; population standard deviation is known to be 5. Use the supplied 95% rule mean ± 1.96 × sigma / sqrt(n). Compute the interval and name one assumption.

Teacher explanation: SE 1, margin 1.96, interval [18.04, 21.96]. Require independent normal sampling and **known sigma**; bias is not repaired by the interval.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: SE 1, margin 1.96, interval [18.04, 21.96]. Require independent normal sampling and **known sigma**; bias is not repaired by the interval. Require intermediate reasoning and the stated scope. Fresh retry: Mean8, n16, **known sigma**4. Expected reasoning: SE1, margin1.96, interval[6.04, 9.96] under the same assumptions. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit research reasoning

> Use saved evidence and assumptions to choose the next question.

Project and next-week bridge

Add a labelled uncertainty intervals **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of uncertainty intervals.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Effect size and practical importance. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Delayed return and support record

Suggested return points in this level: ai-7/13, ai-7/15, ai-7/18. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of uncertainty intervals

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

Before class, solve the supported practice: Assume 16 independent normal measurements with observed mean 10 and known population standard deviation 2. Use a 95% interval: mean ± 1.96 × sigma / sqrt(n). Compute the **standard error**, margin and interval. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: SE = 2/4 = 0.5; margin = 0.98; interval [9.02, 10.98]. Under assumptions, the method covers the fixed population mean in approximately 95% of repeated samples.

Check every learner before discussion: Use the worked case for Uncertainty intervals. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A narrow interval can be precisely biased; sampling independence and representativeness must be justified.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Use a supplied interval [-1.4,0.3] instead. Require a **prediction** and a reason before comparing.

Expected result: It includes zero and positive differences, so both no difference and slower B remain compatible with that interval. Reason: The new upper endpoint crosses zero.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

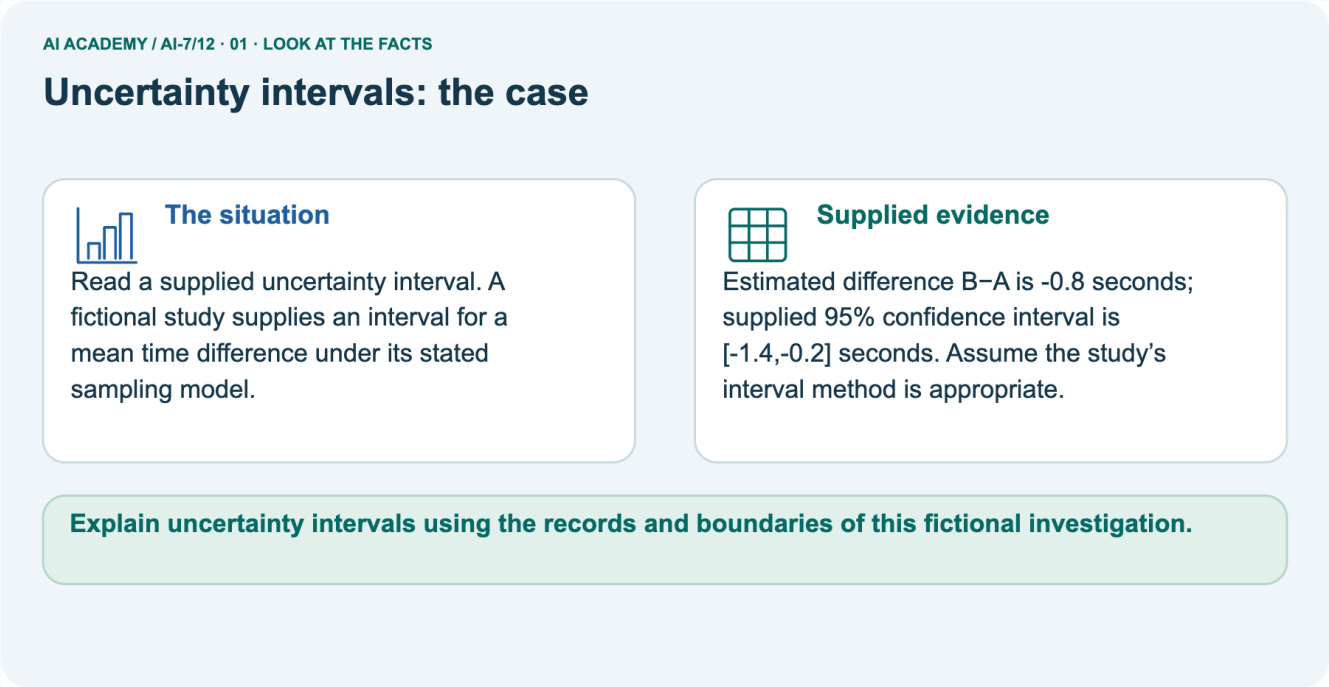
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Uncertainty intervals: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

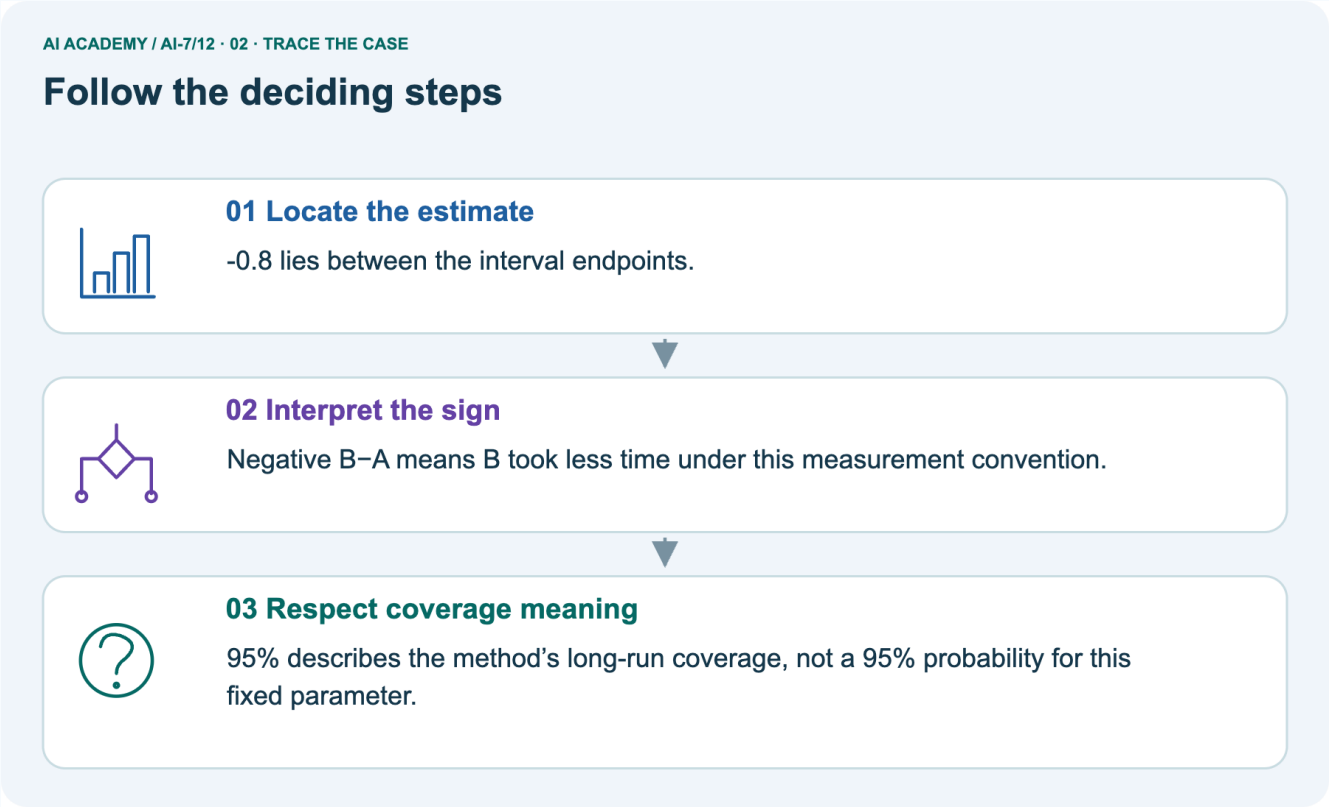
Read a supplied uncertainty interval. A fictional study supplies an interval for a mean time difference under its stated sampling model.

Supplied facts: Estimated difference B–A is -0.8 seconds; supplied 95% **confidence interval** is [-1.4,-0.2] seconds. Assume the study's interval method is appropriate.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



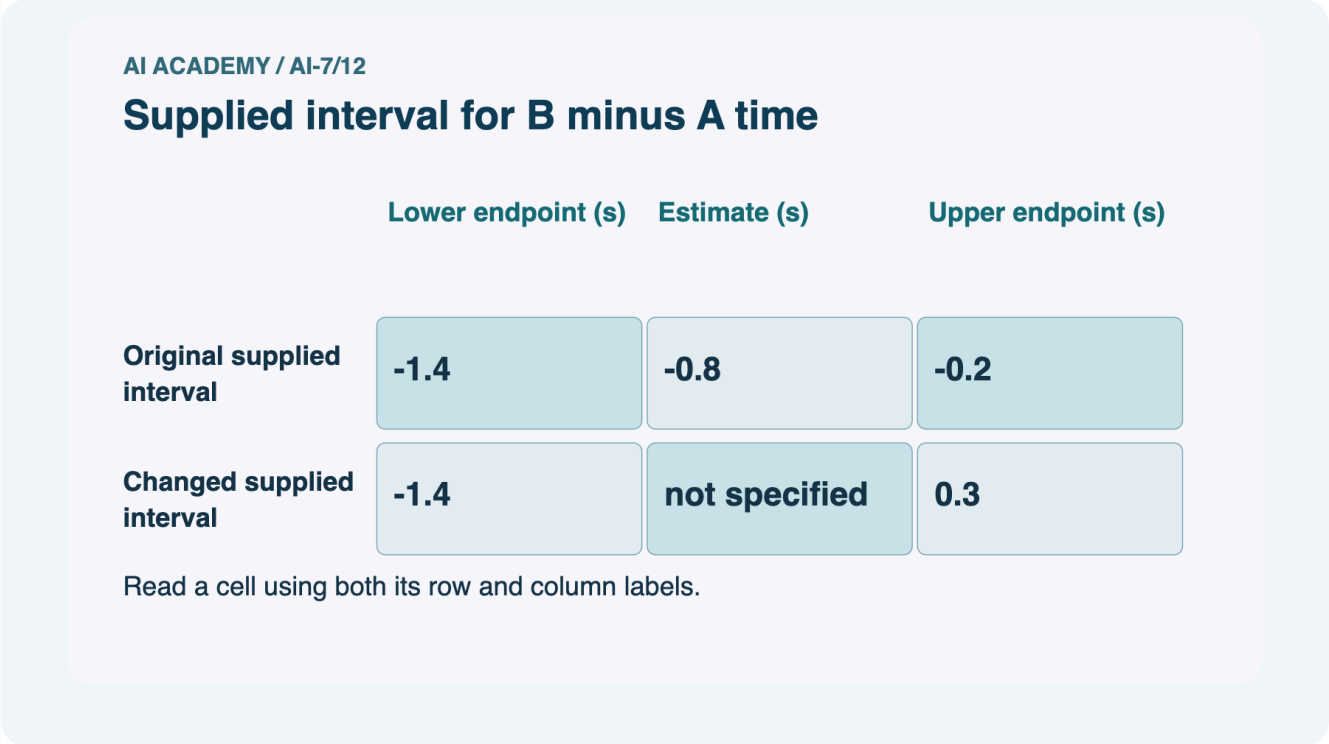
Follow the labelled picture with the matching explanation. Open the vector diagram

Locate the estimate: -0.8 lies between the interval endpoints. → Interpret the sign: Negative B–A means B took less time under this measurement convention. → Respect coverage meaning: 95% describes the method's long-run coverage, not a 95% probability for this fixed parameter.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Supplied interval for B minus A time



Follow the labelled picture with the matching explanation. Open the vector diagram

Supplied interval for B minus A time. Original supplied interval: Lower endpoint (s) = -1.4, Estimate (s) = -0.8, Upper endpoint (s) = -0.2.

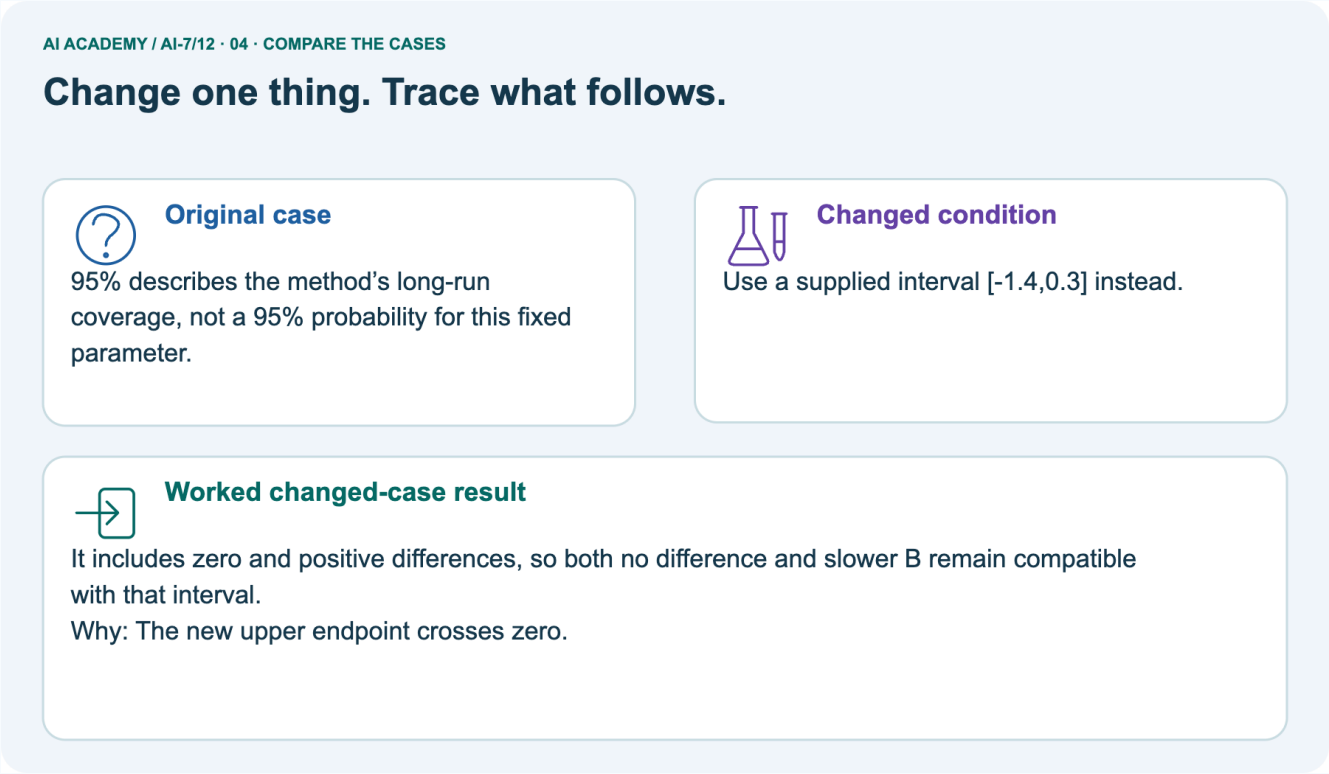
Changed supplied interval: Lower endpoint (s) = -1.4, Estimate (s) = not specified, Upper endpoint (s) = 0.3.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: 95% describes the method’s long-run coverage, not a 95% probability for this fixed parameter. Changed condition: Use a supplied interval [-1.4,0.3] instead.

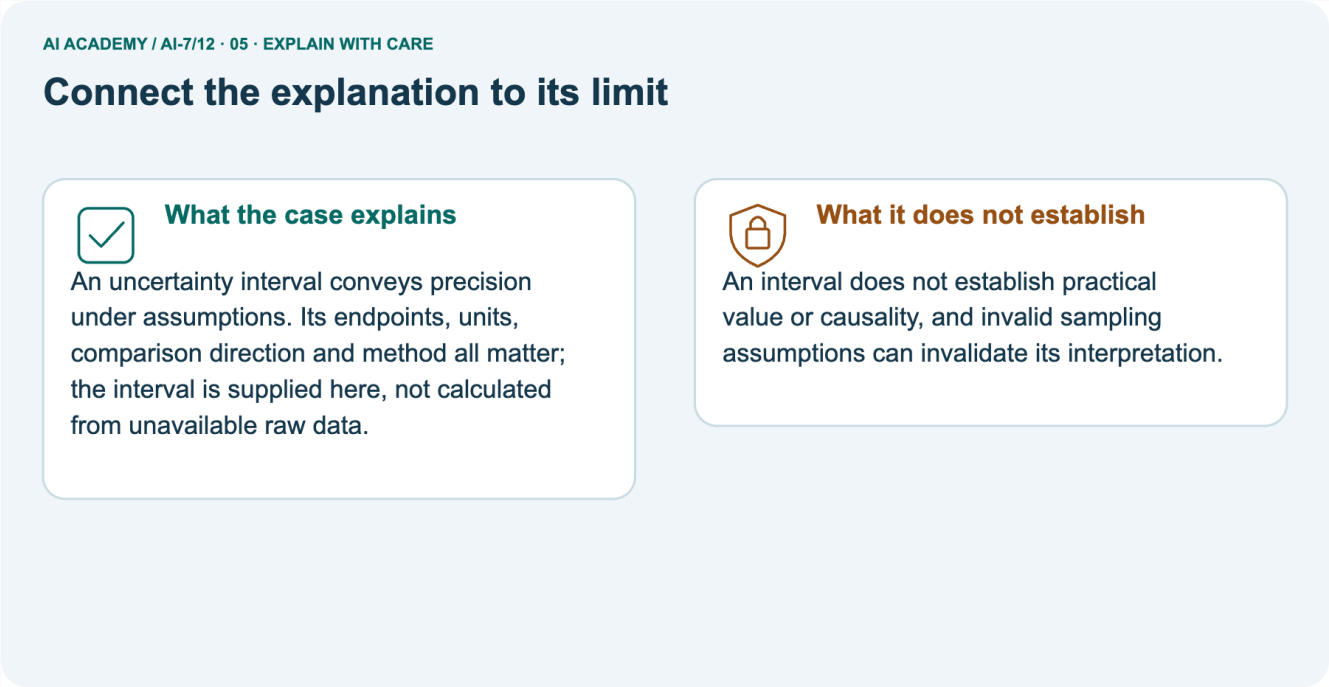
Expected result: It includes zero and positive differences, so both no difference and slower B remain compatible with that interval. Reason: The new upper endpoint crosses zero.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The new upper endpoint crosses zero.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

An uncertainty interval conveys precision under assumptions. Its endpoints, units, comparison direction and method all matter; the interval is supplied here, not calculated from unavailable raw data.

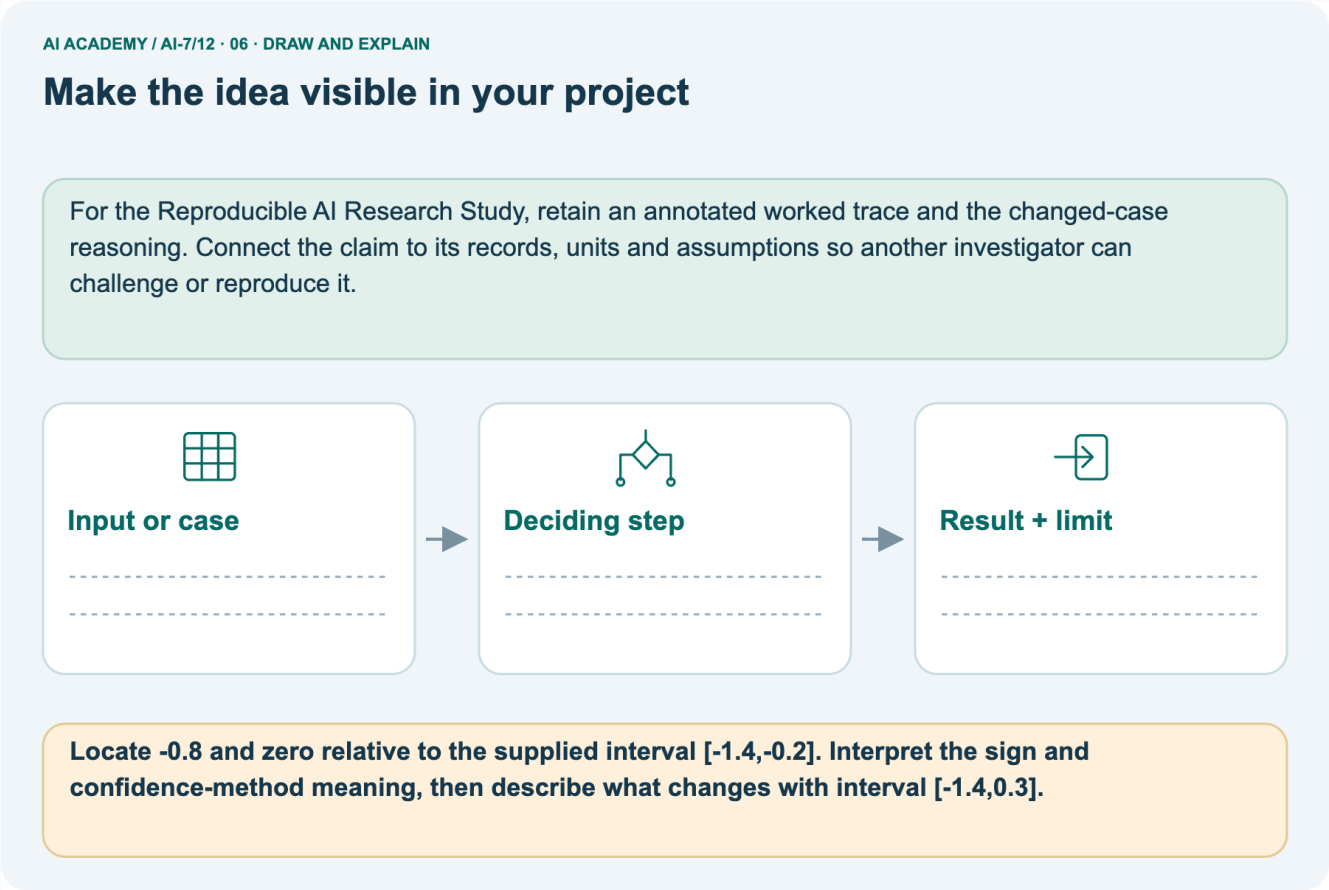
Limit: An interval does not establish practical value or causality, and invalid sampling assumptions can invalidate its interpretation.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: An interval does not establish practical value or causality, and invalid sampling assumptions can invalidate its interpretation.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Locate -0.8 and zero relative to the supplied interval [-1.4,-0.2].

Interpret the sign and confidence-method meaning, then describe what changes with interval [-1.4,0.3].

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: -0.8 lies between the interval endpoints.

Negative B–A means B took less time under this measurement convention. 95% describes the method's long-run coverage, not a 95% probability for this fixed parameter.

Changed case: It includes zero and positive differences, so both no difference and slower B remain compatible with that interval. The new upper endpoint crosses zero.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Uncertainty intervals: the case

02. Follow the deciding steps

03. Supplied interval for B minus A time

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.